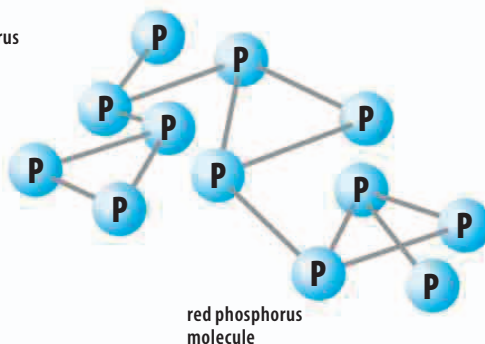
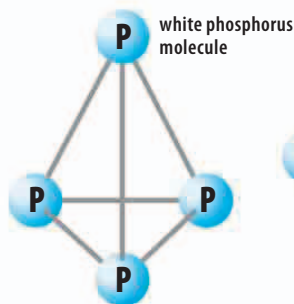
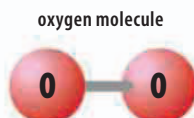


Molecular elements

Atoms get involved in reactions and form molecules to become more stable. So even when they are pure, most elements do not exist as single unbonded atoms. However, the molecules they form consist of only one type of atom.

▷ Increased stability

The oxygen in the air exists as a molecule of two oxygen atoms (O_2). When bonded together, the oxygen atoms are in a more stable state.

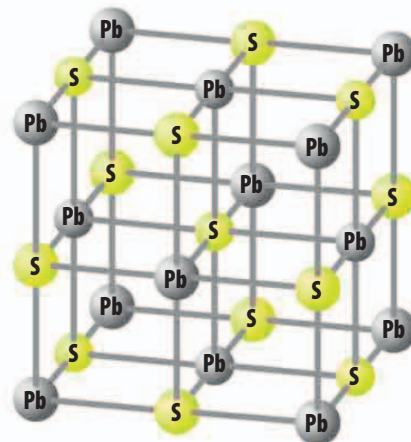


△ Allotropes

Some elements can form more than one shape of molecule and each form is called an allotrope. Phosphorus (P) has red and white allotropes. The red allotrope has stronger bonds and is more stable than the white one, which reacts very easily—even burning on contact with air.

Crystals

A crystal forms when the large numbers of molecules of an element or a compound are all joined together in a repeating pattern. For example, a diamond is a form of carbon made up of repeating tetrahedral units.

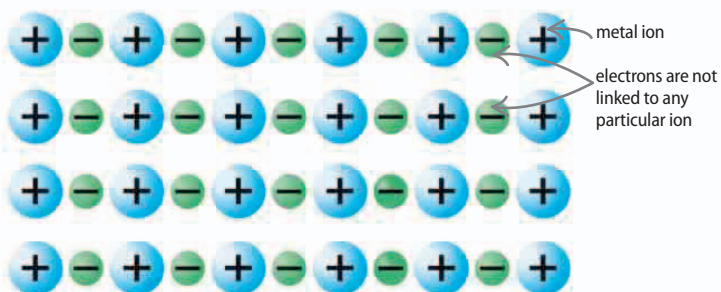


△ Galena

Galena is a compound of lead (Pb) and sulfur (S). Its formula is simply PbS and the lead and sulfur atoms form a cube of atoms. A galena crystal is made up of these cubic units.

Metallic bonds

Metal atoms lose their outer electrons easily, forming positively charged ions (see page 112) surrounded by a sea of shared electrons. The attraction between the negatively charged electrons and positively charged ions creates metallic bonds that “glue” the structure together.



▽ Strong material

The free electrons are shared by all neighboring ions and can slide past each other. This means that metal objects can be deformed, or bent, without breaking.

